INCH-POUND

MIL-C-48670 (AR)
AMENDMENT 4
29 MAY 1992
SUPERSEDING
AMENDMENT 3
23 FEBRUARY 1989

MILITARY SPECIFICATION

CHARGE ASSEMBLY FOR M207E5 OR M207E6 WARHEADS INERT PARTS, FOR

This Amendment forms a part of Military Specification MIL-W-48670 (AR) dated 21 March 1986 and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

In the specification wherever "M207E5" appears add "or M207E6."

PAGE 1

2.1.1, specifications, Military: Add the following:

"MIL-C-48659 - Charge Assembly for M207E5 Warhead Loading, Assembling and Packing"

PAGE 2

2.2, Add: "ASTM E 348 Microhardness of Materials"

PAGE 3

3.5.1, Delete paragraph in its entirety and substitute the following:

"3.5.1 Process documentation. Concurrent with submission of the First Article Sample (see paragraph 4.3) the contractor shall submit a written description of the manufacturing process used for the body and liner. As a minimum, the process description shall include intermediate steps in press operations and specifics of the heat treatment process used together with the means by which the process can be verified in production. The method of verification shall be reviewed by and must be acceptable to the Government. The contractor's written plan of process and process controls shall be prepared in accordance

AMSC N/A FSC 1336

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with Data Item Description DI-P-1604 (Tailored). Proposed changes to the process which produced the accepted First Article sample shall be submitted to the Government prior to actual implementation of the change. A new First Article Sample may be required for each proposed process change as determined by the Government."

PAGE 4

4.3.1, After first sentence, add the following:

"The first article sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packing shall be the same as used in regular production and shall be obtained from the same source of supply. A first Article Sample of portion thereof, as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity or quality as determined by the contracting officer (See 3.5.1)."

4.3.1, Liner quantity: Delete "20" and substitute "30." Plate quantity: Delete "20" and substitute "10." Pad, compression quantity: Delete "20" and substitute "10."

4.3.2, Delete paragraph in its entirety and add the following:

"4.3.2 Testing

4.3.2.1 Starting material. Sample material of the starting stock for the liner, shall be selected prior to fabrication and used for hardness, chemical composition and grain size tests.

4.3.2.2 Finished machine liners. Ten finished machine liners from a single annealed batch shall be used for testing for grain size and hardness.

4.3.2.3 Static penetration. Twenty finished machined liners will be delivered to a designated Government facility for loading and testing in accordance with 4.4.3.2. This test is for information only."

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4.3.5 Delete paragraph in its entirety and substitute the following:

"4.3.5 Completion. Upon completion of the first article testing, sample disposition procedure shall be in accordance with 6.5."

PAGE 5

Delete Note 1 in its entirety and substitute the following:

"1. Upon completion of Examination of Defects select ten (10) of the twenty (20) liners for hardness of the finished liner. These same liners shall then be used for grain size of the finished liner."

Table I

Liner: Examination of defects, delete "20" and substitute "30"

Liner: Add "Penetration (see note 2) 20 3.1 4.4.3.2"

Plate: examination of defects, delete "20" and substitute "10"

Pad Compression: examination of defects, delete "20" and substitute "10"

add Note: "2. For informational purposes only."

PAGE 6

4.4.1, Delete the first sentence in its entirety and substitute the following:

"Inspection lots for the liner, body, plate, PIC, and compression pad shall be identified with a lot number in accordance with provisions of MIL-A-48078 and MIL-STD-1168. The contractor shall request the initial lot interfix number from the government procuring activity.

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PAGE 7 & 8

* Delete 4.4.2.1 in its entirety and substitute included revised 4.4.2.1.

PAGE 9 & 9a

* Delete 4.4.2.2 in its entirety and substitute included revised 4.4.2.2

PAGE 10

Delete 4.4.2.3 in its entirety and substitute included revised 4.4.2.3.

PAGE 11

Delete 4.4.2.4 in its entirety and substitute included revised 4.4.2.4.

PAGE 12

Delete 4.4.2.5 in its entirety and substitute included revised 4.4.2.5.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48670 (AR)
Amendment 4

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9387148
4.4.2.1	Liner				NEXT HIGHER ASSEMBLY 9399673
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter (O.D)		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Total runout of O.D. with datums A and B		0.40%	3.2	Gage
104	Depth of datum A from base		0.40%	3.2	Gage
105	Length of datum A to datum B		0.40%	3.2	Gage
106	Inside spherical radius		0.40%	3.2	Gage
107	Wall thickness at apex end (See note 7 on dwg.)		0.40%	3.2	Gage
108	Wall thickness between datum A and datums B		0.40%	3.2	Gage
109	Perpendicularity of base of liner with datums A and B		0.40%	3.2	Gage
110	Straightness between datum A and datum B (See note 5 on dwg.)		0.40%	3.2	Gage
111	Length from datum A to inside spherical radius		0.40%	3.2	Gage
112	Wall thickness variation between datum A and datum B (See note 4 on dwg.)		0.40%	3.2	Gage

REMARKS

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.1	Liner (continued)		9387148
			NEXT HIGHER ASSEMBLY 9399673

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
113	Hardness of starting stock (See note 1)	Mat'l Sample		3.2	4.6.1
114	Chemical composition of starting stock (See note 1)	Mat'l Sample		3.2	4.6.2
115	Grain size of starting stock (See note 1)	Mat'l Sample		3.2	4.6.3
116	Hardness of finished liner (See note 11B on dwg.)	4.4.3.1.1		3.2	4.6.4
117	Grain size of finished liner (See note 11A on dwg.)	4.4.3.1.1		3.2	4.6.5
118	Evidence of discontinuity of machined surface (See note 10 on dwg.)		0.40%	3.2	Visual
119	Presence of four slots on O.D.		0.40%	3.2	Visual
Minor					
201	63 Finish (See note 6 on dwg.)		0.65%	3.2	Gage
202	Workmanship		0.65%	3.6	Visual

NOTES:

1. Starting stock - See note 2 of drawing.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48670 (AR)
Amendment 4

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Body		9387149
			NEXT HIGHER ASSEMBLY 9399673

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wall thickness		0.40%	3.2	Gage
102	Total length		0.40%	3.2	Gage
103	Large inside diameter (I.D) Note 1		0.40%	3.2	Plug gage
104	Middle I.D.		0.40%	3.2	Gage
105	True position of middle I.D.		0.40%	3.2	Gage
106	Small I.D.		0.40%	3.2	Gage
107	Total runout of small I.D		0.40%	3.2	Gage
108	Depth of middle I.D. to small I.D.		0.40%	3.2	Gage
109	Small outside diameter (O.D.)		0.40%	3.2	Gage
110	Runout small O.D.		0.40%	3.2	Gage
111	Large O.D.		0.40%	3.2	Gage
112	Runout large O.D.		0.40%	3.2	Gage
113	Parallelism of small face to datum B		0.40%	3.2	Gage
114	Perpendicularity of datum B to datum A		0.40%	3.2	Gage
115	Improper taper angle		0.40%	3.2	Gage
117	Improper length to rear basic diameter		0.40%	3.2	Gage

NOTES:

1/ Large inside diameter shall be verified with plug gage at the point approaching the full depth.

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48670 (AR)
Amendment 4

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9387149
4.4.2.2	Body				NEXT HIGHER ASSEMBLY 9399673
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	125 Finish (See note 3 on dwg.)		0.65%	3.2	Gage
202	Protective finish (see note 4 on dwg.)		0.65%	3.2	Visual
203	Workmanship		0.65%	3.6	Visual
204	Improper chamfer at front end		0.65%	3.2	Visual
205	Lot identification illegible		0.65%	3.2	Visual
806a					

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48670 (AR)
Amendment 4

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9387158
4.4.2.3	Plate				NEXT HIGHER ASSEMBLY 9399673
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter		0.40%	3.2	Gage
102	True position of inside diameter		0.40%	3.2	Gage/Visual
103	Large outside diameter		0.40%	3.2	Gage
104	Depth, inside diameter		0.40%	3.2	Gage
Minor					
201	Overall thickness		0.65%	3.2	Gage
202	Presence of chamfer		0.65%	3.2	Gage/Visual
203	Thickness of outside diameter		0.65%	3.2	Gage
204	125 Finish (See note 4 on dwg.)		0.65%	3.2	Gage
205	Protective finish (See note 5 on dwg.)		0.65%	3.2	Visual
206	Workmanship		0.65%	3.6	Visual

NOTES:

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48670 (AR)
Amendment 4

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Body, PIC		9399691
			NEXT HIGHER ASSEMBLY 9399689

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large inside diameter		0.40%	3.2	Gage
102	Small inside diameter		0.40%	3.2	Gage
103	Length large inside diameter		0.40%	3.2	Gage
104	True position large inside diameter		0.40%	3.2	Gage
105	Outside diameter		0.40%	3.2	Gage
106	True position, outside diameter		0.40%	3.2	Gage
Minor					
201	Overall length		0.65%	3.2	Gage
202	Protective finish (See note 4 on dwg.)		0.65%	3.2	Visual
203	Workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48670 (AR)
Amendment 4

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9387189
4.4.2.5	Pad, Compression				NEXT HIGHER ASSEMBLY 9399687
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Material (See note 2 on dwg.)				Certification
Minor					
201	Inside diameter		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Workmanship		0.65%	3.6	Visual

NOTES:

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4.4.3.1.1.b, Delete the paragraph in its entirety and substitute with the following:

"If five consecutive annealed batches meet the heat treatment requirements reduce the sampling to a minimum of one liner per week's production of annealed batches or one liner from the last annealed batch producing a total of no more than 5000 pieces whichever occurs first. Test results must be submitted to and accepted by the government."

4.4.3.2, Add the following:

"4.4.3.2 Static penetration test. Sampling shall be in accordance with the following:

1. Ten (10) tip charge assemblies will be loaded at a designated government facility in accordance with the assembly procedures specified in MIL-C-48659 using liners provided in para. 4.3.1.

2. Static penetration testing of these assemblies will be conducted at a designated government facility using the test methods and target materials as specified in MIL-C-48659.

3. If penetration results of the 10 samples meets the requirements of MIL-C-48659 the remaining liners will be held for future reference.

4. If any sample from step 4 fails the specified penetration requirement, the remaining 10 liners will be loaded and step 2 repeated.

5. All penetration results shall be submitted in accordance with DI-T-5247 (Tailored) (see 6.4)."

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4.6, Delete the paragraph in its entirety and substitute the following:

"4.6 Metallurgical and chemical tests. the following tests shall be performed using equipment and procedures prescribed by the cited ASTM."

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4.6.1, Include the following at the end of the first sentence: "in accordance with ASTM B 248."

4.6.2, Include the following at the end of the first sentence: "in accordance with ASTM b 248."

4.6.3, Include the following at the end of the first sentence: "in accordance with ASTM B248."

* 4.6.4, Delete paragraph in its entirety and substitute the following:

"4.6.4 Hardness of finished liner. The sample liners shall be sectioned longitudinally into four equal parts and marked for exact location relative to the finished liner. The hardness of the finished liner shall be determined in accordance with ASTM-E-384. The microhardness measurements shall be taken at the following locations: 0.10 inch from the base and every 0.7 inch thereafter in a longitudinal direction. For each location three (3) readings shall be taken, for a total of nine (9) readings on each section (See Fig. 1). A total of thirty-six (36) readings shall be taken for the finished liner. The hardness readings shall be identified as to their location on the liner. The recorded readings shall be analyzed for variations in hardness in both the longitudinal and axial directions. If any of the sample liners do not meet the hardness requirements specified on the drawing, the liner heat treat batch represented by the samples shall be rejected. The hardness test data shall be submitted in accordance with DI-T-5247 (Tailored) (See 6.4). This is a destructive test."

4.6.5, Delete the first and second sentence and substitute the following:

"Two quarter test sections, 90 degrees apart, from 4.6.4 shall be utilized for taking grain size. Examination shall be made at the same three locations which were hardness tested (See Figure 1)."

6.3, At end of paragraph delete "Dover, New Jersey 07801-5001" and substitute "Picatinny Arsenal, N.J. 07806-5000."

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6.4, Delete and substitute as follows:

"6.4 Data requirements - Deliverable data required by the specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.3	Ammunition Data	DI-L-1410
3.5.1 & 3.5.2	Manufacturing Methods Report	DI-P-1604 (Tailored) (See note 1)
4.4.3.2, 4.6.1, 4.6.2, 4.6.3, 4.6.4, 4.6.5	Test Report	DI-T-5247 (Tailored)

The above data items shall be delivered to the following:

Commander
U.S. Army Armament Research, Development and Engineering Center
ATTN: SMCAR-QAT-B
Picatinny Arsenal, N.J. 07806-5000

Note 1: One copy shall also be sent to Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-CCH-A, Picatinny Arsenal, NJ 07806-5000."

PAGE 15a

Add new figure included.

PAGE 17

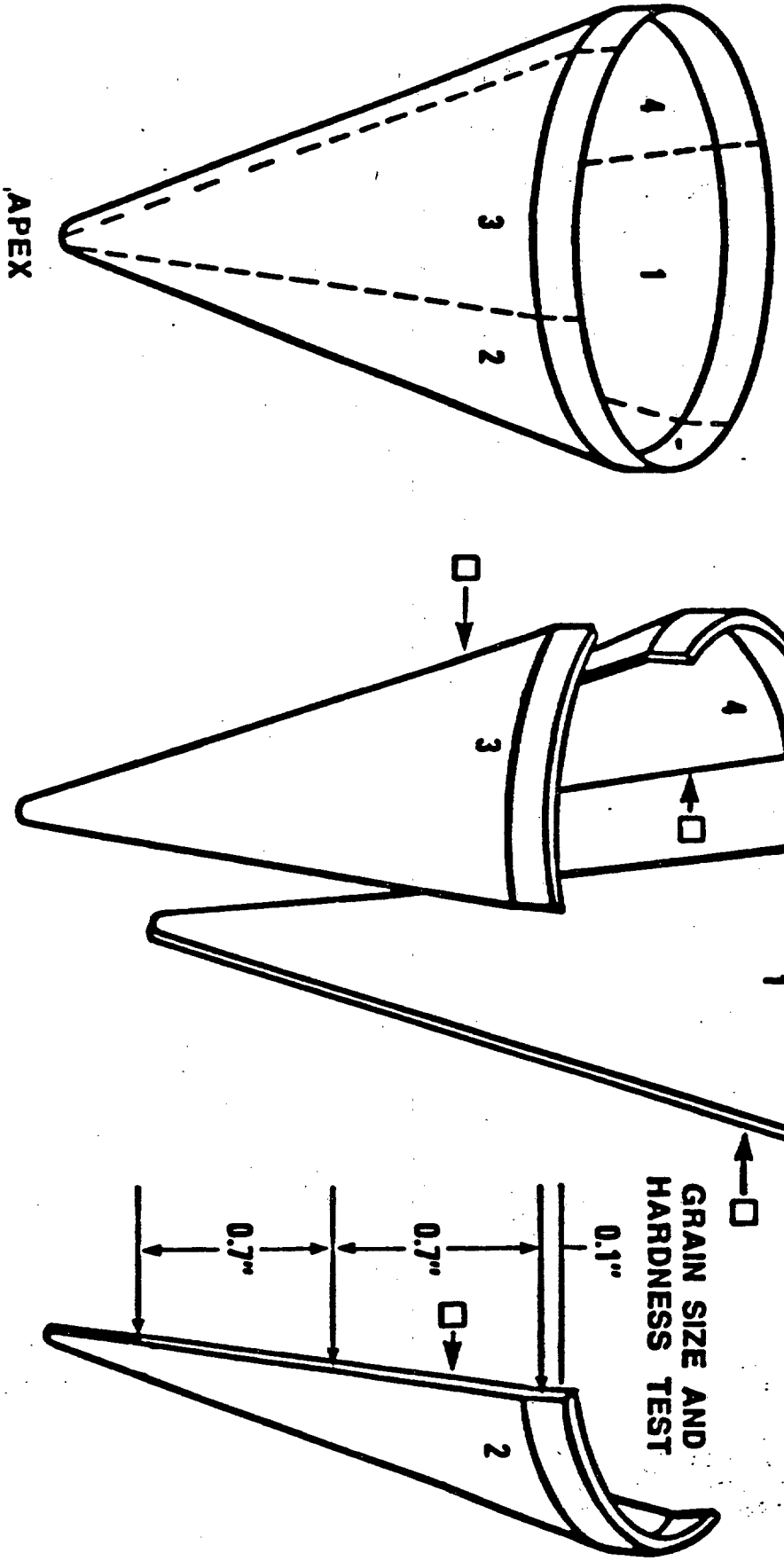
6.5, Add new paragraph as follows:

"6.5, Add new paragraph as follows:

"6.5 Submission of test hardware. Upon completion of First Article Testing, all samples shall be sent to Commander, ARDEC, ATTN: SMCAR-CCH-A, Picatinny Arsenal, NJ 07806-5000."

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SCHEMATIC OF SECTIONED LINER



□→ Designates the cross-section of each required section to be tested.

Figure 1

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